

RESOLUÇÃO DO DESAFIO

```
!pip install mysql-connector-python
```

```
Collecting mysql-connector-python
  Downloading mysql_connector_python-9.7.0-cp312-cp312-manylinux_2_28_x86_64.whl.metadata (11 kB)
  Downloading mysql_connector_python-9.7.0-cp312-cp312-manylinux_2_28_x86_64.whl (21.7 MB)
    21.7/21.7 MB 55.3 MB/s eta 0:00:00
Installing collected packages: mysql-connector-python
Successfully installed mysql-connector-python-9.7.0
```

```
import mysql.connector

conn = mysql.connector.connect(host="35.199.115.174",
                               port=3306,
                               user='loobox-challenge',
                               password='loooq-challenge',
                               database='loobox-challenge')

print('Banco de dados conectado')
```

Banco de dados conectado

```
import mysql.connector
import pandas as pd

query = 'SHOW TABLES;'

df = pd.read_sql(query, conn)
print(df)
```

```
/tmp/ipykernel_2488/146329309.py:7: UserWarning: pandas only supports SQLAlchemy connectable (engine/connection) or database
df = pd.read_sql(query, conn)
Tables_in_loobox-challenge
0      IMDB_movies
1      data_product
2      data_product_sales
3      data_store_cad
4      data_store_sales
```

QUESTÃO 1 - QUAIS SÃO OS 10 PRODUTOS MAIS CAROS DA EMPRESA?

```
query = 'SELECT * FROM data_product ORDER BY PRODUCT_VAL DESC LIMIT 10;'
df = pd.read_sql(query, conn)
print(df)
```

```
/tmp/ipykernel_2488/533567683.py:2: UserWarning: pandas only supports SQLAlchemy connectable (engine/connection) or database
df = pd.read_sql(query, conn)
PRODUCT_COD      PRODUCT_NAME \
0      301409  Whisky Escoces THE MACALLAN Ruby Garrafa 700ml...
1      176185  Whisky Escoces JOHNNIE WALKER Blue Label Garra...
2      315481  Cafeteira Expresso 3 CORACOES Tres Modo Vermelho
3      100280  Vinho Portugues Tinto Vintage QUINTA DO CRASTO...
4      320046  Escova Dental Eletrica ORAL B D34 Professional...
5      190817  Champagne Rose VEUVE CLICQUOT PONSARDIM Garraf...
6      153795  Champagne Frances Brut Imperial MOET Rose Garr...
7      311397  Conjunto de Panelas Allegra em Inox TRAMONTINA...
8      147706  Whisky Escoces CHIVAS REGAL 18 Anos Garrafa 750ml
9      154431  Champagne Frances Brut Imperial MOET & CHANDON...

PRODUCT_VAL  DEP_NAME  DEP_COD  SECTION_NAME  SECTION_COD
0      741.99    BEBIDAS      2      BEBIDAS      4
1      735.90    BEBIDAS      2      BEBIDAS      4
2      499.00    BEBIDAS      2      BEBIDAS      4
3      445.90    BEBIDAS      2      VINHOS      30
4      399.90  PERFUMARIA      8      HIGIENE BUCAL  39
5      366.90  MERCEARIA      6  ARTIGOS-PARA-O-LAR      2
6      359.90  MERCEARIA      6  ARTIGOS-PARA-O-LAR      2
7      359.00  MERCEARIA      6  ARTIGOS-PARA-O-LAR      2
8      329.90    BEBIDAS      2      BEBIDAS      4
9      315.90  MERCEARIA      6  ARTIGOS-PARA-O-LAR      2
```

QUESTÃO 2 - QUAIS SEÇÕES EXISTEM NOS DEPARTAMENTOS "BEBIDAS" E "PADARIA"?

```
query = "SELECT DISTINCT SECTION_NAME, DEP_NAME FROM data_product WHERE DEP_NAME IN ('BEBIDAS', 'PADARIA');"
df = pd.read_sql(query, conn)
print(df)
```

```
/tmp/ipykernel_2488/264300616.py:2: UserWarning: pandas only supports SQLAlchemy connectable (engine/connection) or database
df = pd.read_sql(query, conn)
  SECTION_NAME DEP_NAME
0          BEBIDAS  BEBIDAS
1          VINHOS  BEBIDAS
2  DOCES-E-SOBREMESAS  PADARIA
3    QUEIJOS-E-FRIOS  PADARIA
4          CERVEJAS  BEBIDAS
5          PADARIA  PADARIA
6          REFRESCOS  BEBIDAS
7          GESTANTE  PADARIA
```

QUESTÃO 3 - QUAL O TOTAL DE VENDAS EM \$ DE CADA ÁREA DE NEGÓCIO NO PRIMEIRO TRIMESTRE DE 2019?

```
query = 'SELECT * FROM data_store_cad LIMIT 5;'
df = pd.read_sql(query, conn)
print(df)
```

```
/tmp/ipykernel_2488/500136829.py:2: UserWarning: pandas only supports SQLAlchemy connectable (engine/connection) or database
df = pd.read_sql(query, conn)
  STORE_CODE STORE_NAME  START_DATE  END_DATE BUSINESS_NAME BUSINESS_CODE
0          1  Sao Paulo  2006-10-01          Varejo          1
1          2   Chicago  2007-10-01          Varejo          1
2          3     Roma   2008-10-01          Varejo          1
3          4    Tokio   2009-10-01          Varejo          1
4          5     Paris  2019-01-01  Proximidade          2
```

```
query = """
SELECT sc.BUSINESS_NAME, SUM(ps.SALES_VALUE) AS TOTAL_VENDAS
FROM data_product_sales ps
JOIN data_store_cad sc ON ps.STORE_CODE = sc.STORE_CODE
WHERE ps.DATE BETWEEN '2019-01-01' AND '2019-03-31'
GROUP BY sc.BUSINESS_NAME;
"""
df = pd.read_sql(query, conn)
print(df)
```

```
/tmp/ipykernel_2488/2308394804.py:8: UserWarning: pandas only supports SQLAlchemy connectable (engine/connection) or database
df = pd.read_sql(query, conn)
  BUSINESS_NAME  TOTAL_VENDAS
0        Varejo   81032347.65
1        Farma   81776691.73
2      Atacado  80384884.60
3        Posto  32072326.40
4  Proximidade  80171122.80
```

CASO 1 - FILTRO DE PESQUISA PARA ITENS DE LISTA:

```
def retrieve_data(product_code, store_code, date):

    if isinstance(product_code, int):
        product_code = [product_code]
    if isinstance(store_code, int):
        store_code = [store_code]

    product_str = ', '.join(str(x) for x in product_code)
    store_str = ', '.join(str(x) for x in store_code)

    query = f"""
SELECT * FROM data_product_sales
WHERE PRODUCT_CODE IN ({product_str})
AND STORE_CODE IN ({store_str})
AND DATE BETWEEN '{date[0]}' AND '{date[1]}'
"""

    df = pd.read_sql(query, conn)
    return df
```

```
pesquisa = retrieve_data([18, 48, 172], [1, 2, 3, 4], ['2019-01-01', '2019-03-31'])
print(pesquisa)
```

```
/tmp/ipykernel_2488/1122101381.py:18: UserWarning: pandas only supports SQLAlchemy connectable (engine/connection) or database
df = pd.read_sql(query, conn)
  STORE_CODE  PRODUCT_CODE  DATE  SALES_VALUE  SALES_QTY
```

0	1	18	2019-01-01	708.50	65.0
1	1	18	2019-01-02	1297.10	119.0
2	1	18	2019-01-03	1144.50	105.0
3	1	18	2019-01-04	1090.00	100.0
4	1	18	2019-01-05	893.80	82.0
...	...	...	...	...	...
1075	4	172	2019-03-27	873.08	292.0
1076	4	172	2019-03-28	699.66	234.0
1077	4	172	2019-03-29	669.76	224.0
1078	4	172	2019-03-30	568.10	190.0
1079	4	172	2019-03-31	472.42	158.0

[1080 rows x 5 columns]

CASO 2 - QUERY's ENVIADAS E VIZUALIZAÇÃO DESTAS:

```
query1 = """
SELECT STORE_CODE, STORE_NAME, START_DATE, END_DATE, BUSINESS_NAME, BUSINESS_CODE
FROM data_store_cad
"""
```

```
query2 = """
SELECT STORE_CODE, DATE, SALES_VALUE, SALES_QTY
FROM data_store_sales
WHERE DATE BETWEEN '2019-01-01' AND '2019-12-31'
"""
```

```
df1 = pd.read_sql(query1, conn)
df2 = pd.read_sql(query2, conn)
```

```
print(df1.head())
print(df2.head())
```

```
/tmp/ipykernel_2488/2802442013.py:12: UserWarning: pandas only supports SQLAlchemy connectable (engine/connection) or database
df1 = pd.read_sql(query1, conn)
/tmp/ipykernel_2488/2802442013.py:13: UserWarning: pandas only supports SQLAlchemy connectable (engine/connection) or database
df2 = pd.read_sql(query2, conn)
```

	STORE_CODE	STORE_NAME	START_DATE	END_DATE	BUSINESS_NAME	BUSINESS_CODE
0	1	Sao Paulo	2006-10-01		Varejo	1
1	2	Chicago	2007-10-01		Varejo	1
2	3	Roma	2008-10-01		Varejo	1
3	4	Tokio	2009-10-01		Varejo	1
4	5	Paris	2019-01-01		Proximidade	2

	STORE_CODE	DATE	SALES_VALUE	SALES_QTY
0	1	2019-01-01	196623.22	12838
1	10	2019-01-01	126795.44	4933
2	11	2019-01-01	223937.00	7724
3	12	2019-01-01	200251.80	7043
4	13	2019-01-01	196623.22	12838

```
import pandas as pd
```

```
df_merged = pd.merge(df1, df2, on='STORE_CODE')
```

```
df_merged['DATE'] = pd.to_datetime(df_merged['DATE'])
```

```
df_filtrado = df_merged[(df_merged['DATE'] >= '2019-10-01') & (df_merged['DATE'] <= '2019-12-31')]
```

```
print(df_filtrado.head())
```

	STORE_CODE	STORE_NAME	START_DATE	END_DATE	BUSINESS_NAME	BUSINESS_CODE	\
273	1	Sao Paulo	2006-10-01		Varejo	1	
274	1	Sao Paulo	2006-10-01		Varejo	1	
275	1	Sao Paulo	2006-10-01		Varejo	1	
276	1	Sao Paulo	2006-10-01		Varejo	1	
277	1	Sao Paulo	2006-10-01		Varejo	1	

	DATE	SALES_VALUE	SALES_QTY
273	2019-10-01	187601.54	12160
274	2019-10-02	332666.14	21643
275	2019-10-03	253401.57	16489
276	2019-10-04	257709.59	16724
277	2019-10-05	216934.36	14120

```
df_tm = df_filtrado.groupby(['STORE_NAME', 'BUSINESS_NAME']).apply(
    lambda x: round(x['SALES_VALUE'].sum() / x['SALES_QTY'].sum(), 2)
).reset_index()
```

```
df_tm.columns = ['Loja', 'Categoria', 'TM']
```

```
df_tm = df_tm.sort_values('Loja').reset_index(drop=True)
```

```
print(df_tm)
```

	Loja	Categoria	TM
0	Bahia	Atacado	15.39
1	Bangkok	Posto	13.67
2	Belem	Proximidade	15.37
3	Berlin	Proximidade	15.39
4	Buenos Aires	Atacado	15.39
5	Chicago	Varejo	15.53
6	Dubai	Atacado	15.39
7	Hong Kong	Farma	26.35
8	London	Farma	28.99
9	Madri	Farma	29.03
10	Miami	Posto	13.67
11	New York	Proximidade	15.39
12	Paris	Proximidade	15.39
13	Rio de Janeiro	Farma	29.59
14	Roma	Varejo	15.39
15	Salvador	Atacado	15.39
16	Sao Paulo	Varejo	15.39
17	Sidney	Posto	13.67
18	Tokio	Varejo	15.39
19	Vancouver	Posto	13.67

/tmp/ipykernel_2488/1169831630.py:2: DeprecationWarning: DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated. Use .apply() instead.
df_tm = df_filtrado.groupby(['STORE_NAME', 'BUSINESS_NAME']).apply(

```
import matplotlib.pyplot as plt
```

```
import matplotlib.pyplot as plt
```

```
fig, ax = plt.subplots(figsize=(6, 7))
ax.axis('off')
```

```
tabela = ax.table(
    cellText=df_tm.values,
    colLabels=df_tm.columns,
    cellLoc='center',
    loc='center'
)
```

```
tabela.auto_set_font_size(False)
tabela.set_fontsize(9)
tabela.scale(1.2, 1.4)
```

```
plt.title('TM por Loja e Categoria - Q4 2019', pad=20)
plt.tight_layout()
plt.savefig('caso2.png', dpi=150, bbox_inches='tight')
plt.show()
```

TM por Loja e Categoria — Q4 2019

Loja	Categoria	TM
Bahia	Atacado	15.39
Bangkok	Posto	13.67
Belem	Proximidade	15.37
Berlin	Proximidade	15.39
Buenos Aires	Atacado	15.39
Chicago	Varejo	15.53
Dubai	Atacado	15.39
Hong Kong	Farma	26.35
London	Farma	28.99
Madri	Farma	29.03
Miami	Posto	13.67
New York	Proximidade	15.39
Paris	Proximidade	15.39
Rio de Janeiro	Farma	29.59
Roma	Varejo	15.39
Salvador	Atacado	15.39
Sao Paulo	Varejo	15.39
Sidney	Posto	13.67
Tokio	Varejo	15.39
Vancouver	Posto	13.67

CASO 3 - GRÁFICO IMDB

```
query = "SELECT * FROM IMDB_movies LIMIT 5;"
df_imdb = pd.read_sql(query, conn)
print(df_imdb.columns.tolist())
```

```
/tmp/ipykernel_2488/503488121.py:2: UserWarning: pandas only supports SQLAlchemy connectable (engine/connection) or database
df_imdb = pd.read_sql(query, conn)
['Id', 'Title', 'Genre', 'Director', 'Actors', 'Year', 'Runtime', 'Rating', 'Votes', 'RevenueMillions', 'Metascore']
```

```
query = """
SELECT Title, Rating, RevenueMillions, Metascore
FROM IMDB_movies
WHERE RevenueMillions IS NOT NULL
AND Metascore IS NOT NULL;
"""
df_imdb = pd.read_sql(query, conn)
print(df_imdb.head())
```

```
/tmp/ipykernel_2488/2655144856.py:7: UserWarning: pandas only supports SQLAlchemy connectable (engine/connection) or databas
df_imdb = pd.read_sql(query, conn)
```

	Title	Rating	RevenueMillions	Metascore
0	Guardians of the Galaxy	8.0	333.0	76
1	Prometheus	7.0	126.0	65
2	Split	7.0	138.0	62
3	Sing	7.0	270.0	59
4	Suicide Squad	6.0	325.0	40

```
import matplotlib.pyplot as plt

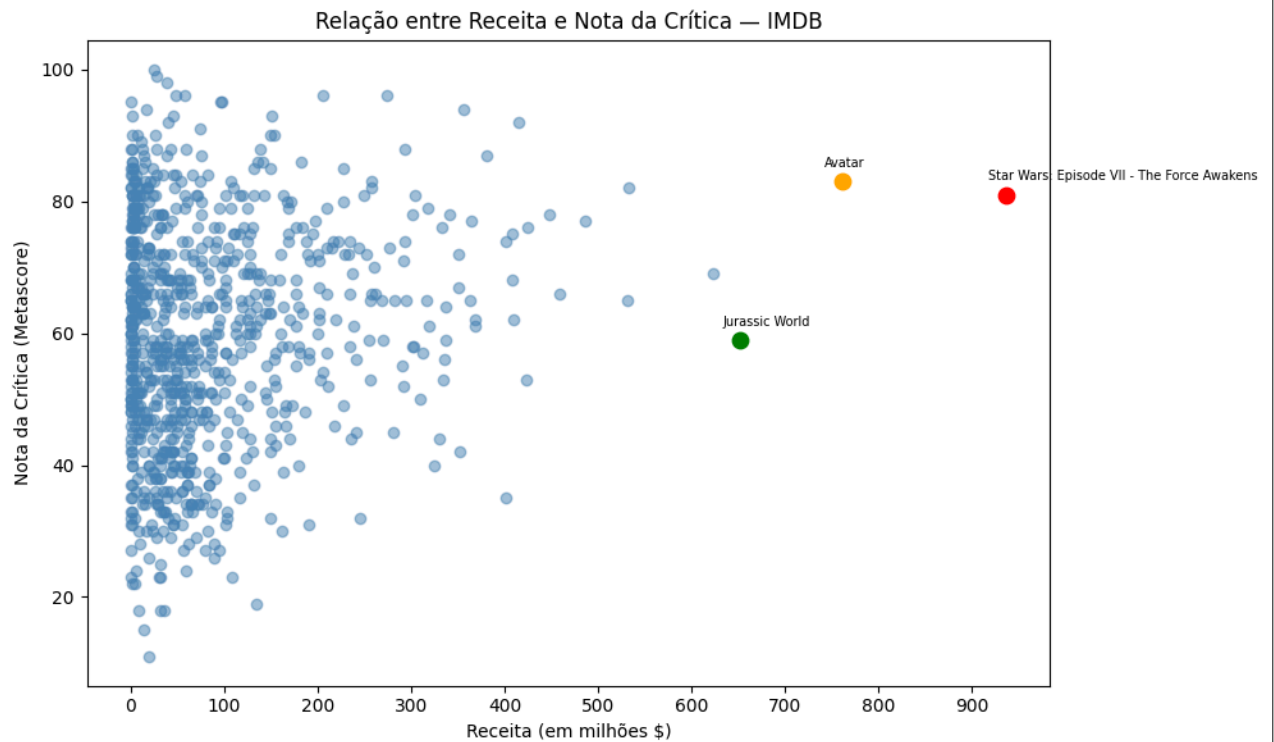
fig, ax = plt.subplots(figsize=(10, 6))

ax.scatter(df_imdb['RevenueMillions'], df_imdb['Metascore'], alpha=0.5, color='steelblue')

# destaca os 3 maiores
destaques = df_top.values
cores = ['red', 'orange', 'green']
for i, (title, revenue, meta) in enumerate(destaques):
    ax.scatter(revenue, meta, color=cores[i], zorder=5, s=80)
    ax.annotate(title, (revenue, meta), textcoords="offset points", xytext=(-10, 8), fontsize=7)

ax.set_xlabel('Receita (em milhões $)')
```

```
ax.set_ylabel('Nota da Crítica (Metascore)')  
ax.set_title('Relação entre Receita e Nota da Crítica — IMDB')  
ax.set_xticks([0, 100, 200, 300, 400, 500, 600, 700, 800, 900])  
  
plt.tight_layout()  
plt.savefig('caso3.png', dpi=150, bbox_inches='tight')  
plt.show()
```



Optei por gerar um gráfico de dispersão com o objetivo de avaliar a correlação entre a nota da crítica e a receita de bilheteria. Uma das hipóteses era verificar se filmes bem avaliados necessariamente gerariam maior receita.